A Teen Scientist's Warning: Caffeine Could Quiet Crucial Brain Genes in Kids

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You probably know someone who can't start the day without coffee or an energy drink. Caffeine is everywherein soda, tea, even chocolate. But Sophia Zeng, a 16-year-old from Tsinghua International School in Beijing, says we might be too casual about its effects on young brains. Her research, presented at the 2026 Regeneron International Science & Engineering Fair (ISEF), suggests caffeine could alter the activity of genes crucial for brain development, and that the impact may be more severe in kids than in adults.

Sophia's curiosity began with a personal observation. Her mom felt focused after drinking tea, but Sophia often got headaches when she tried to quit. "If I stopped drinking it, I'd receive headaches," she says. That led her to a bigger question: Why does caffeine affect people so differently?

She started with **bioinformatics**analyzing published data to spot new patterns. Her analysis of studies on adult mice hinted that caffeine could change which genes in the brain were expressed, or turned on. To test this, she moved to the lab. Sophia grew rat brain cells in petri dishes, exposed them to different caffeine levels, and then examined them under a microscope using **fluorescent dyes**. The results were striking: dishes with more caffeine had fewer neurons, and those neurons had fewer **synapses**the critical connections where brain cells communicate. "Caffeine seemed to make neuronal networks simpler and scarcer," she explains.

But Sophia didn't stop at cells in a dish. She turned to young zebrafish, a common model for studying development. Fish exposed to caffeine swam more and made **erratic choices**, venturing into areas of the tank that calm fish usually avoidclassic signs of anxiety. When she measured the activity of four genes involved in brain development, all four were less active in the caffeinated fish. Intriguingly, some of these gene changes were not seen in the adult rat neurons, suggesting that caffeine may affect the brain differently depending on age.

Together, the mouse data, rat cell experiments, and fish studies point to a consistent pattern: caffeine seems to turn down genes that help build and maintain brain connections. This could impair both brain development and behavior. Sophia believes we need greater **skepticism** about caffeine as a harmless tool. "People tend to think about caffeine as a helpful tool for focus," she notes, "and dismiss potential downsides." But her findings highlight **trade-offs**and those **trade-offs** may harm young brains more than we know. With many kids consuming caffeine daily, her work is a wake-up call to reconsider what's in that soda can.

Word Treasure

synapses -- The connections between brain cells where messages are sent and received.

bioinformatics -- Using computer programs to analyze large amounts of biological data, like gene information.

fluorescent dyes -- Special colorful chemicals that glow under a microscope to help scientists see tiny parts of cells.

erratic choices -- Unpredictable and unusual decisions, like a fish swimming into places it normally would avoid.

skepticism -- A questioning attitude; not automatically believing something is safe or true.

trade-offs -- The compromises or losses that come with gaining something else, like sacrificing health for focus.

Background you need

Caffeine is a stimulant that affects the central nervous system, and it is the most widely consumed psychoactive substance in the world.

Adolescents are often more sensitive to caffeine than adults because their brains are still developing, and their bodies metabolize it more slowly.

The Regeneron International Science & Engineering Fair (ISEF) is the world's largest pre-college science competition, held annually in the United States, where thousands of high school students present original research.

Zebrafish (Danio rerio) are small fish commonly used in laboratories to study genetics and development because they share about 70% of their genes with humans and grow quickly.

Break it down

LEAD: Caffeine is everywhere, but a teen scientist questions its safety for young brains.

+- SPECIFIC: Sophia Zeng observed that quitting caffeine gave her headaches, sparking her curiosity.

KEY RESEARCH: Bioinformatics analysis of mouse studies hinted that caffeine could change which genes are turned on in the brain.

+- LAB WORK: Rat brain cells exposed to caffeine had fewer neurons and fewer synapses -- the connections where cells communicate.

+- FISH STUDIES: Zebrafish exposed to caffeine swam more, made anxious choices, and had reduced activity of four brain-development genes.

FINDINGS: Caffeine seems to turn down genes that help build and maintain brain connections, with effects that may differ by age.

+- CAVEAT: Some gene changes in fish were not seen in adult rat neurons, suggesting age-dependent vulnerability.

IMPLICATION: Greater skepticism is needed; caffeine may harm young brains more than adults, especially with frequent consumption.

Why it matters

Many 12-14 year-olds consume caffeine daily in sodas and energy drinks; this research suggests it could interfere with brain development during a critical period, potentially affecting learning, mood, and behavior.

Test what you remember

1. **What did Sophia Zeng observe about her own experience with caffeine?**
   - Ⓐ She felt more focused like her mom.
   - Ⓑ She often got headaches when she tried to quit.
   - Ⓒ She drank coffee every morning without problems.
   - Ⓓ She found caffeine had no effect on her.
2. **What did Sophia find when she grew rat brain cells in petri dishes with more caffeine?**
   - Ⓐ The cells multiplied faster.
   - Ⓑ There were fewer neurons and fewer synapses.
   - Ⓒ The cells grew larger than normal.
   - Ⓓ No change was observed.
3. **In the zebrafish experiments, what behavior did caffeinated fish show?**
   - Ⓐ They swam more slowly and calmly.
   - Ⓑ They swam more and made erratic choices, showing anxiety.
   - Ⓒ They completely stopped moving.
   - Ⓓ They stayed in one area of the tank.
4. **How many genes involved in brain development did Sophia measure in the fish?**
   - Ⓐ Two
   - Ⓑ Three
   - Ⓒ Four
   - Ⓓ Six
5. **What did some of the gene changes in fish show that was different from adult rat neurons?**
   - Ⓐ The same changes occurred.
   - Ⓑ The changes were much smaller.
   - Ⓒ Some gene changes were not seen in adult rat neurons.
   - Ⓓ The fish genes became more active.
6. **According to Sophia, what do many people think about caffeine?**
   - Ⓐ It is harmful to everyone.
   - Ⓑ It is only found in coffee.
   - Ⓒ It is a helpful tool for focus.
   - Ⓓ It has no effect on the brain.

Answer key (try the quiz first!): 1: B 2: B 3: B 4: C 5: C 6: C

Questions to think about

1. **Positive / Benefit: If confirmed, this research could lead to stricter regulations on caffeine in products aimed at children and teenagers, protecting adolescent brain health.**

2. **Negative / Critical: The study was done in lab animals (mice, rat cells, fish), not humans, so the direct effects on kids are still uncertain. More research is needed before drawing firm conclusions.**

3. **Neutral / Analytical: Caffeine's effects are known to be dose-dependent and vary by age; this study adds evidence that young brains are more vulnerable, consistent with known differences in how children and adults process caffeine.**

4. **Forward-looking: Policymakers and educators may need to reconsider the availability of energy drinks in schools, and teens should be informed about potential risks to their developing brains.**

MY THOUGHTS (at least 20 words):